

Enoch Zhu

(682) 895-9331 | enoch.zhu154@gmail.com | github.com/Wavefire5201 | linkedin.com/in/enochzhu1 | enochzhu.com

EDUCATION

The University of Texas at Austin

Austin, TX

B.S. Computer Science, Minor in Statistics and Data Science, Certificate in Japanese

Aug 2024 — May 2028

- Relevant Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Statistical Machine Learning, Geospatial Artificial Intelligence

TECHNICAL SKILLS

- **Infrastructure & DevOps:** Linux, Docker, Nginx, Traefik, Cloudflare DNS, Tailscale, WireGuard, GitHub Actions, CI/CD
- **Programming Languages:** Python, TypeScript, JavaScript, SQL, Bash, C, C++, Java, Go, Rust, Swift, R
- **Backend & Data:** PostgreSQL, PostGIS, pgvector, pgTAP, REST APIs, Supabase, FastAPI, Hono, Stripe Connect, Firebase
- **AI/ML:** RAG, hybrid retrieval (pgvector HNSW + RRF), cross-encoder reranking, tool-calling agents, MCP, embeddings
- **Certifications:** CompTIA A+ (2023), GIAC Foundational Cybersecurity Technologies (2024)
- **Languages:** Mandarin Chinese (fluent), Japanese (intermediate)

EXPERIENCE

Icelis Global

Jun 2025 — Aug 2025

Software Engineering Intern

Doha, Qatar (Remote)

- Automated **70%** of an ESG sustainability reporting workflow, using LlamaIndex extraction into typed Postgres
- Shipped a Firebase time-tracking app for **50+ users**, using server-authoritative timestamps and 4-role approvals
- Cut manual regulatory search **40%** across **1,076 pages**, achieved by a RAG chatbot with page-level citations

Sustainable Building Initiative (SBI)

Aug 2024 — Aug 2026

Director of External Technologies

Austin, TX

- Led a **12-person** engineering team across **7 repositories** serving **75+ members** and a **\$200M+** project portfolio
- Built the member portal in **Next.js**, **FastAPI**, and Postgres: auth, document storage, and an AI agent chat
- Ranked portal document search by fusing vector and keyword hits with **RRF**, then a cross-encoder reranker
- Removed a per-query sequential scan without re-embedding, achieved by truncating **4096-dim** vectors to fit HNSW
- Hardened multi-tenant **Row Level Security** on the portal database, using **209** adversarial pgTAP assertions in CI
- Built a Discord meeting bot in Python that records voice channels, mixes per-speaker audio, and transcribes

Bell Flight

Aug 2023 — May 2024

Software Engineering Intern

Fort Worth, TX

- Cut reindexing time **90%** on a legacy .NET and Apache Solr app, achieved by migrating to multi-threaded indexing
- Eliminated manual handling of **1,000+** customer access records, achieved by scripting the updates in Python

Freelance

2021 — Present

Web Developer

Austin, TX (Remote)

- Shipped **5** client sites from first requirements call to launch, using **Astro** and Netlify with custom domains
- Cut page weight with build-time image pipelines, shipping contact forms, sitemaps, and LocalBusiness schema

PROJECTS

Private Cloud Infrastructure | Linux, Docker, Nginx, Cloudflare, Tailscale, WireGuard

Jun 2024 — Present

- Hosted and maintained the SBI production backend for **75+ members** and the Cuaya API, using self-hosted Dokploy
- Ran **19 containers** and **7 health-checked services** on Linux behind an Nginx reverse proxy with automated TLS
- Kept internal services at zero public ingress, achieved by routing them over a Tailscale and WireGuard mesh
- Monitored every host and container across **4TB+** of storage, using Netdata metrics and per-service latency checks

Cuaya | TypeScript, Hono, Drizzle, PostgreSQL, PostGIS, Docker, Traefik

Jul 2026 — Present

- Cut provider search from **1s to 7ms** over **2.9M** US healthcare records, using PostGIS KNN with a query-plan test
- Streamed an **11 GB** federal healthcare provider export into Postgres in 10k-row batches, re-geocoding changes
- Guarded irreversible deletes behind a self-disarming sweep, backed by **350 tests** with cross-user isolation
- Backed up Postgres nightly on self-hosted Linux and drilled a full restore to matching row counts in Docker

Bell Pepper | TypeScript, Supabase, PostgreSQL, PL/pgSQL, Stripe Connect, Expo

Sep 2025 — Present

- Built a **3-sided** farm-to-consumer delivery marketplace for farms, drivers, and customers in one Expo app
- Prevented oversells at checkout across multiple farms, using row locks and server-side price re-reads per order
- Kept payout retries from double-paying farms, using a lease-based queue and per-leg Stripe idempotency keys
- Gated customer addresses behind run assignment: drivers see **~1 km**-rounded coordinates until they claim a run